Business Report -PGP-DSGA

Austo Motor Company

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Austo Automobile

Introduction:

Austo Motor Company is a leading car manufacturer specializing in SUV, Sedan, and Hatchback models. In its recent board meeting, concerns were raised by the members on the efficiency of the marketing campaign currently being used. The board decides to rope in an analytics professional to improve the existing campaign.

Data Description

Variable	Definition
Age	Age of the individual (years)
Gender	Gender (Male or Female or Unknown)
Profession	Occupation or profession
Marital_status	Marital status (Married & Single)
Education	Educational qualification as Graduate and Post Graduate
No_of_Dependents	The number of dependents (e.g., children, elderly parents) that the individual supports financially
Personal_loan	The individual has taken a personal loan "Yes" or "No"
House_loan	Whether the individual has taken a housing loan "Yes" or "No"
Partner_working	The individual's partner is employed "Yes" or "No"
Salary	The individual's salary or income
Partner_salary	The salary or income of the individual's partner
Total_salary	Combination of salary of the individual and their partner salary
Price	Price of a product or service
Make	Type of automobile such as SUV, Sedan, and Hatchback

Understanding the shape of the dataset

Shape means rows and columns in dataset, So There are 1581 rows and 14 columns.

Finding datatypes of Columns present in Dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1581 entries, 0 to 1580
Data columns (total 14 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Age               1581 non-null   int64  
 1   Gender            1528 non-null   object 
 2   Profession        1581 non-null   object 
 3   Marital_status    1581 non-null   object 
 4   Education         1581 non-null   object 
 5   No_of_Dependents  1581 non-null   int64  
 6   Personal_loan     1581 non-null   object 
 7   House_loan        1581 non-null   object 
 8   Partner_working   1581 non-null   object 
 9   Salary            1581 non-null   int64  
 10  Partner_salary    1475 non-null   float64
 11  Total_salary      1581 non-null   int64  
 12  Price             1581 non-null   int64  
 13  Make              1581 non-null   object 
dtypes: float64(1), int64(5), object(8)
memory usage: 173.1+ KB
```

- Dataset contains 1581 entries of 14 columns.
- All the columns present having Non-Null values except Gender and Partner_salary.
- Total_salary is the sum of Partner_salary and Salary.
- It contains int64, object and float64 datatypes for the given columns.
- Age, No_of_Dependents, Salary, Partner_salary, Total_salary and Price columns having statistical data and rest of the columns having categorical data.
- So, we have enough datatypes needed for statistical and categorical analysis of dataset.

Statistical Summary of Dataset

Now we'll check statistical summary of dataset

	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	Price
count	1581.000000	1581.000000	1581.000000	1581.000000	1581.000000	1581.000000
mean	31.922201	2.457938	60392.220114	19233.776091	79625.996205	35597.722960
std	8.425978	0.943483	14674.825044	19670.391171	25545.857768	13633.636545
min	22.000000	0.000000	30000.000000	0.000000	30000.000000	18000.000000
25%	25.000000	2.000000	51900.000000	0.000000	60500.000000	25000.000000
50%	29.000000	2.000000	59500.000000	25100.000000	78000.000000	31000.000000
75%	38.000000	3.000000	71800.000000	38100.000000	95900.000000	47000.000000
max	54.000000	4.000000	99300.000000	80500.000000	171000.000000	70000.000000

Here is the statistical summary of all the columns having statistical values, summary comprises fully populated data present except Partner_salary missing (106) values.

- The average(mean) age is **31.9 years**, with the minimum and maximum being **22 years** and **54 years**, respectively.
- The average(mean) number of dependents is **2.45**, indicating most individuals have **2–3 dependents**.
- The **average(mean) salary** is approximately **60,000**, while the **average(mean) partner salary** is around **19,000**.
- The **combined average(mean) total income** is roughly **80,000**, with the highest reaching **1,71,000**.
- The **average(mean) purchase price of vehicle** is about **36,000**, and the **maximum price** is **70,000**.
- All the amount mentioned in dataset is in Dollars ($)

Missing Values in dataset

	0
Age	0
Gender	53
Profession	0
Marital_status	0
Education	0
No_of_Dependents	0
Personal_loan	0
House_loan	0
Partner_working	0
Salary	0
Partner_salary	106
Total_salary	0
Price	0
Make	0

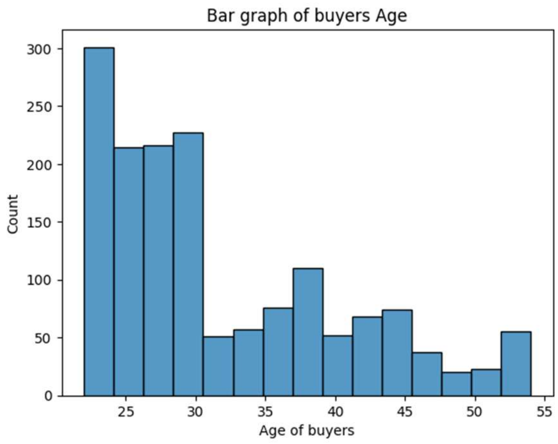
dtype: int64

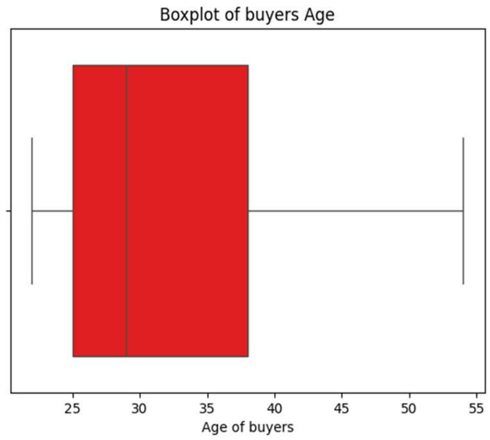
We found missing values in Gender column and Partner_salary column total of 53 and 106 entries respectively.

- Missing values for Gender column are treated by imputation techniques and filled with 'Unknown' value.
- Missing values for Partner_salary column are treated by formulating Imputation techniques.
- No Duplicate records were identified except Gender column is having spelling mistakes, treated by replacing with correct ones.

Univariate Analysis for Numerical Variables

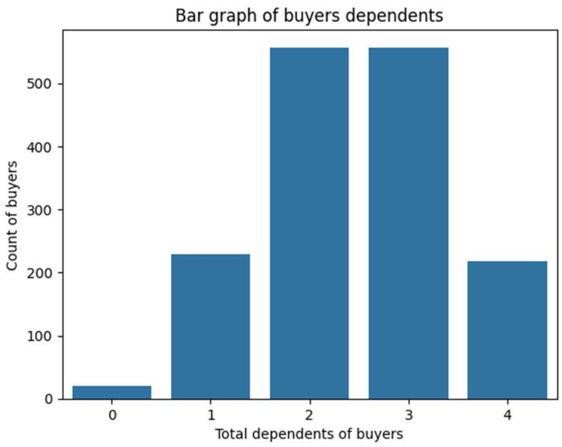
1. Age

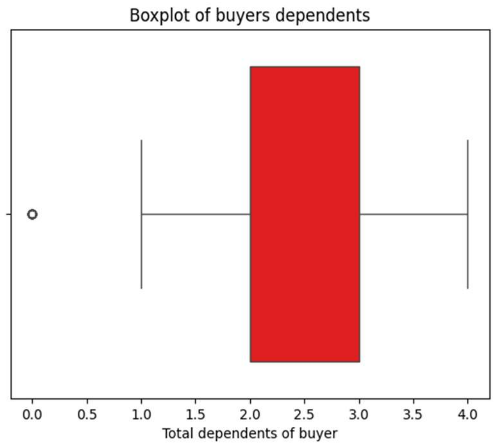




- Majority of buyers are young individuals slab below 30 years of age.
- Shows right skewed distribution with median age is 29 with minimum at 22 and maximum at 54 years.
- 50% of buyers age lie between 25-38 years.

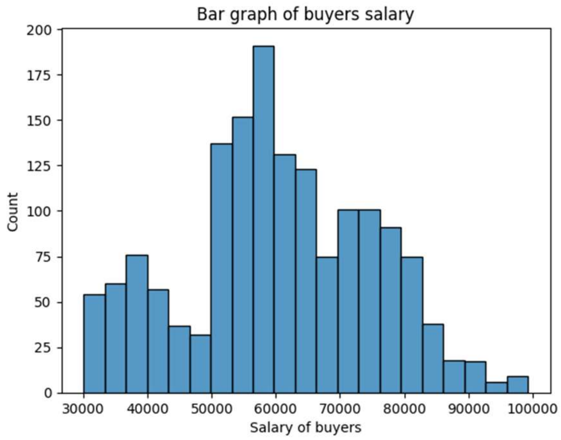
2. No_of_Dependents

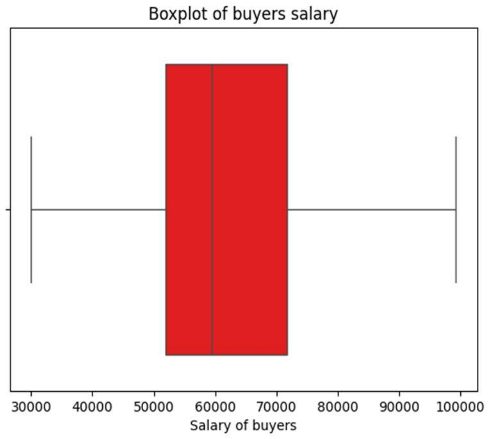




- Majority of buyer having 2-3 dependents.
- Few buyers having NO or 0 dependent shows as the outliers.
- While number of dependents ranges between 0-4.

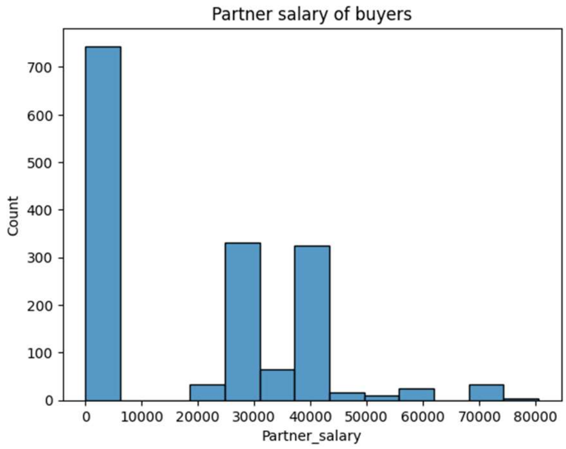
3. Salary

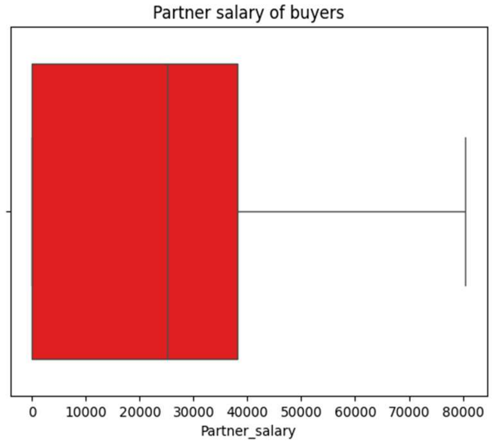




- The Salary ranges between $30000 to $99300 shows significant variation of earnings.
- $59500 will be shown as the median among these.
- While majority of buyers earn between $52000 to $72000.

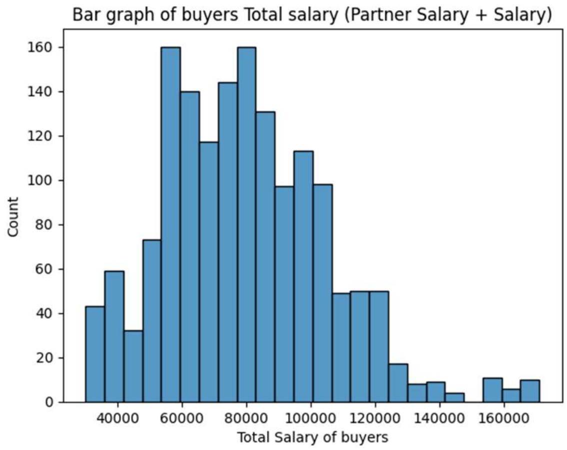
4. Partner_salary

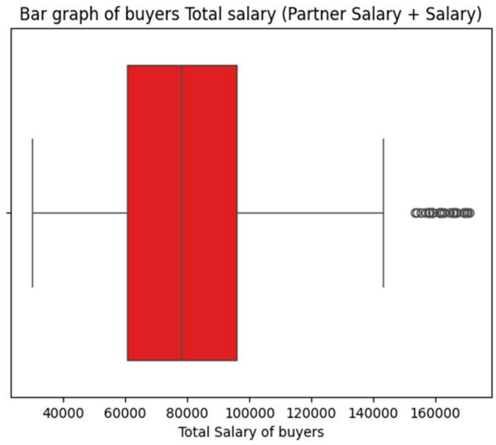




- Median Partner_salary reaches to around $25000.
- While most of the buyers declared 0 or NO partner salary.
- Partner_salary ranges maximum up to $80500 but in a very small proportion.

5. Total_salary

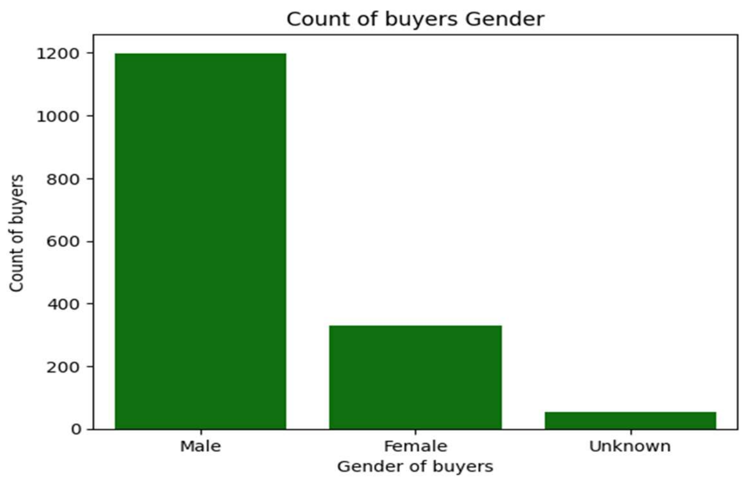




- Total_salary distribution is highly Right skewed.
- Also having outliers present due to extremely higher total salary.
- Total_salary = Salary + Partner_salary
- Median Total_salary is $78000 with most earners are ranges between $60000 and $100000.
- Salaries ranges from $30000 at the lower threshold to $171000 at the upper end, with scattered high-income outliers exceeding $140000.

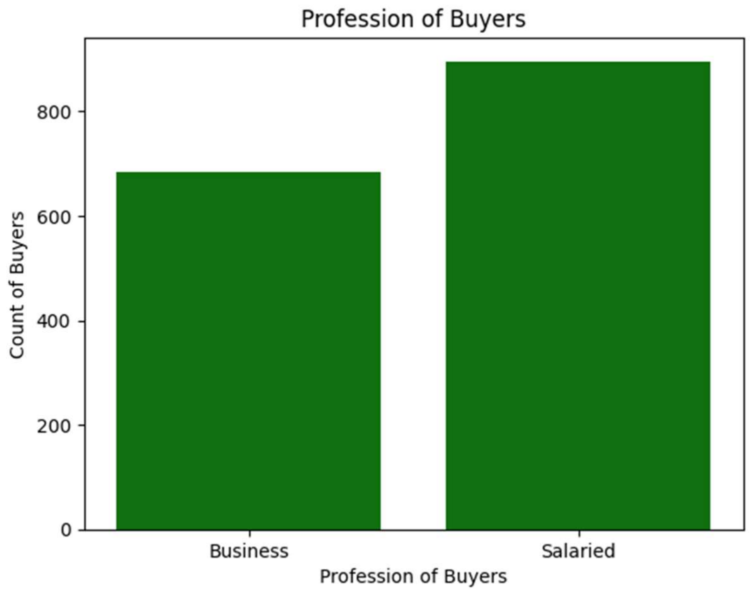
UNIVARIATE ANALYSIS FOR CATEGORICAL VARIABLE DATA:

1. **Gender**



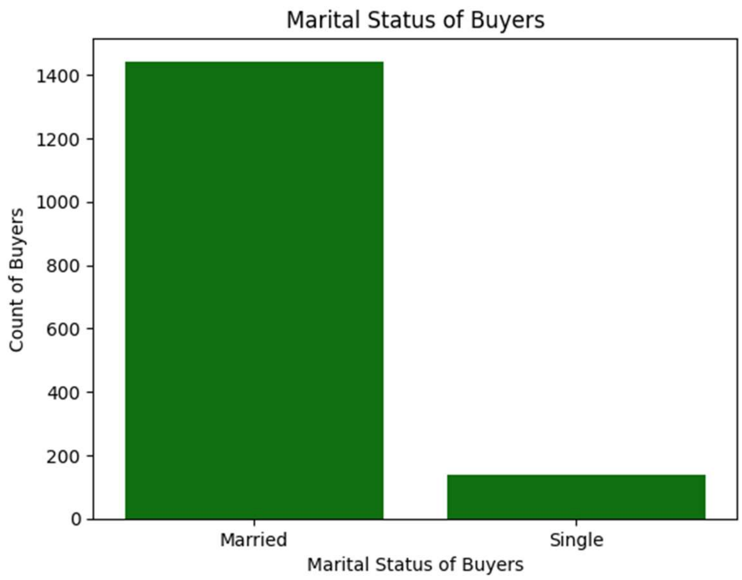
- Majority of buyers are males followed by females.
- Unknown category denotes about missing data that will be either male or female.
- There are 1199 Males, 329 Females and rest 53 are Unknown.

2. Profession



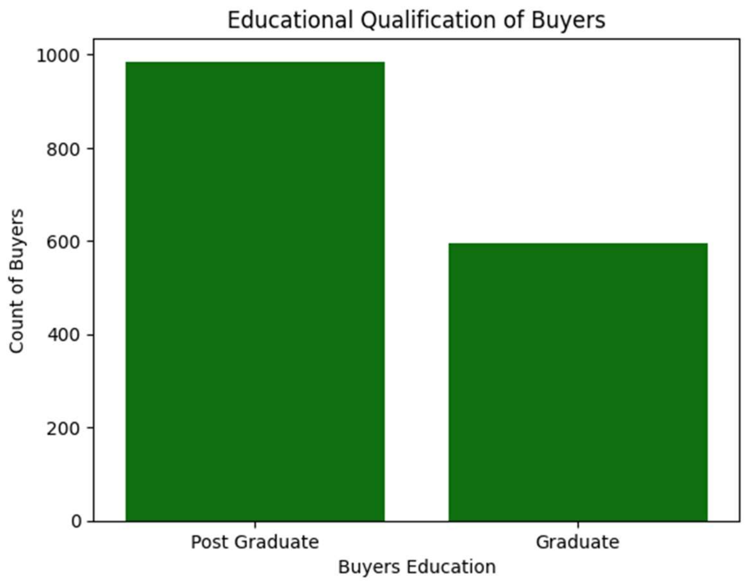
- Majority of buyers are salaried as compare to business profession.

3. Marital Status



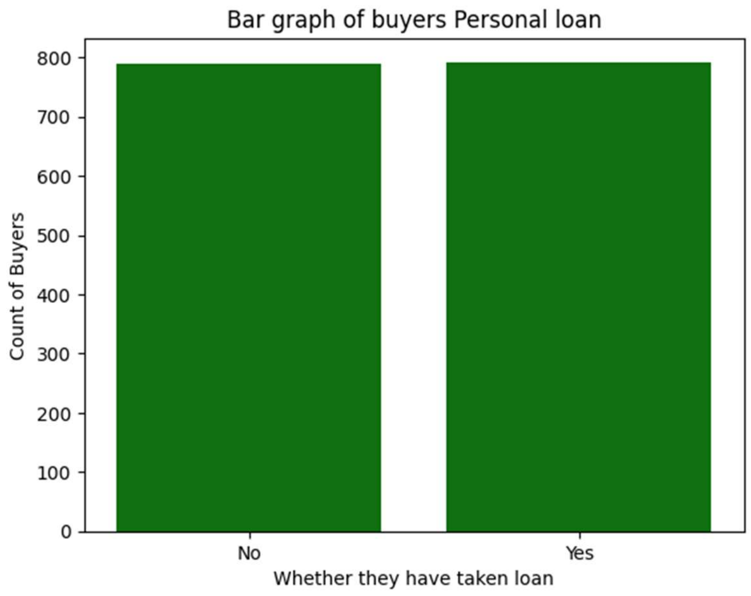
- Around 90% of buyers are Married.

4. Education



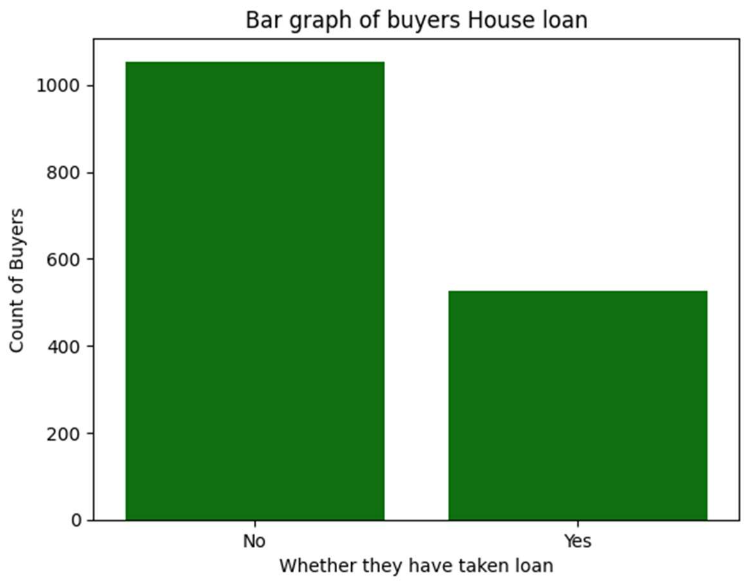
- More buyers are Post Graduate in education.

5. Personal Loan



- Personal Loan are equally split as buyers taken loan is equal to buyers who haven't taken loan.

6. House Loan

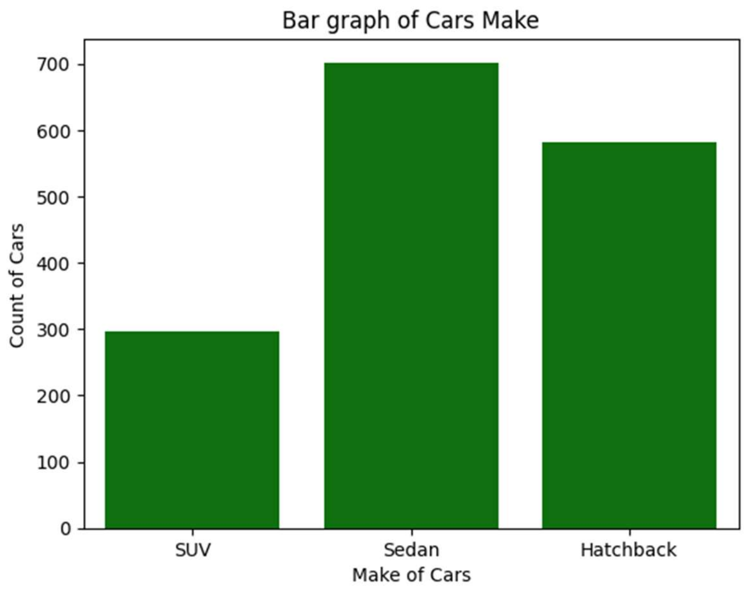


- Most of the people haven't taken House loan.

7. Partner Working

- Partner working are around 850 while not working are around 700, So its having very slight difference.

8. Make



- Sedan is the make preferred among buyers followed by Hatchback and SUV.

OBSERVATION AND INSIGHTS:

<u>INSIGHTS:</u>

- More buyers are salaried than in business.
- The majority of buyers are married.
- Most of the buyers are postgraduates.
- Personal loans are evenly split-about half have one, half don't.
- Majority of buyers don't have a house loan; fewer than half do.
- More buyers having working partners.
- Sedans are the most popular car type, followed by hatchbacks, then SUVs.

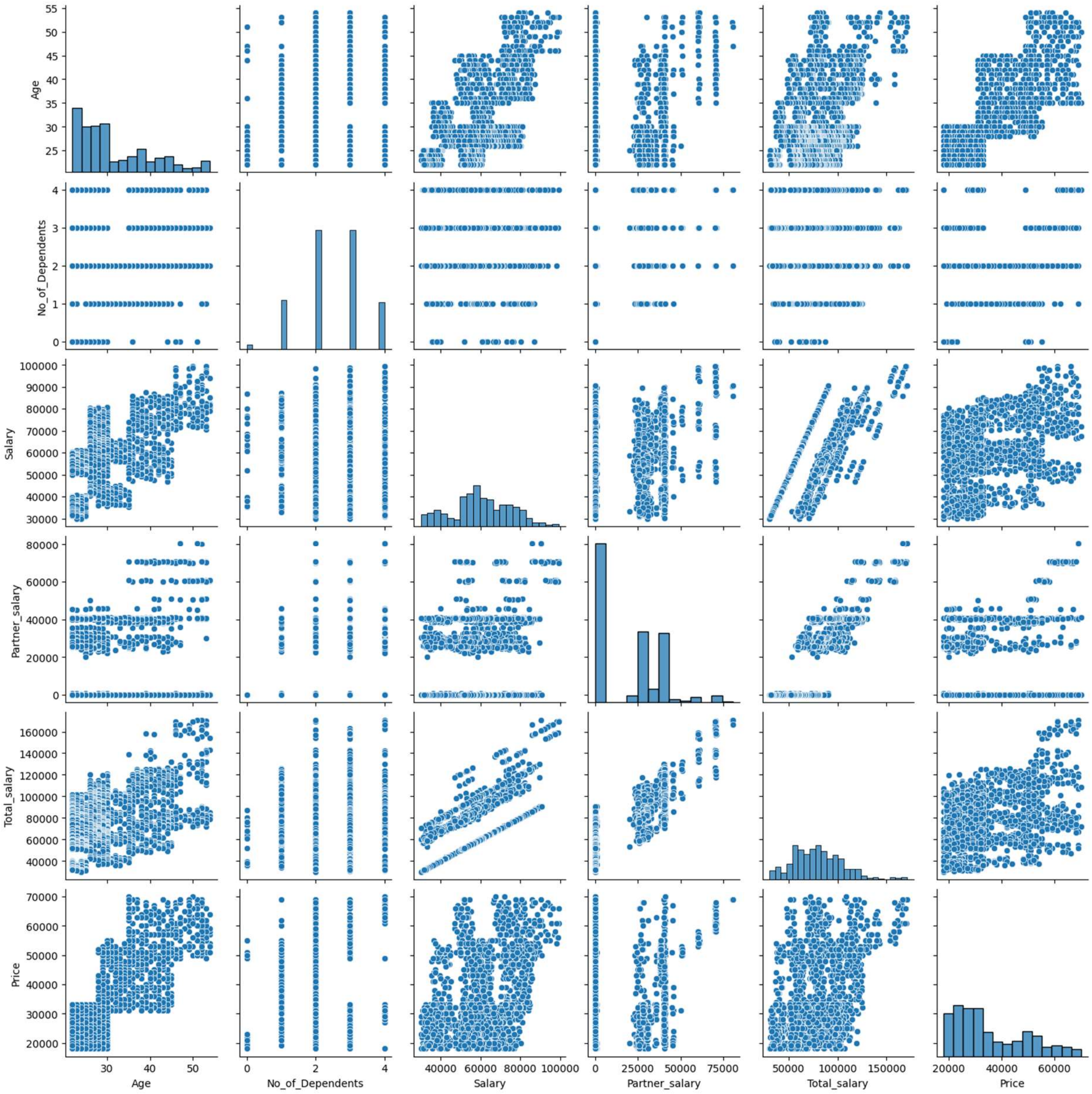
<u>Observations</u>

- Most number of buyers are salaried shows stable income source that leads to purchase of vehicles.
- Most of the young age buyers prefer mid-range cars as align with salary.
- Small number of customers prefer high-range cars.

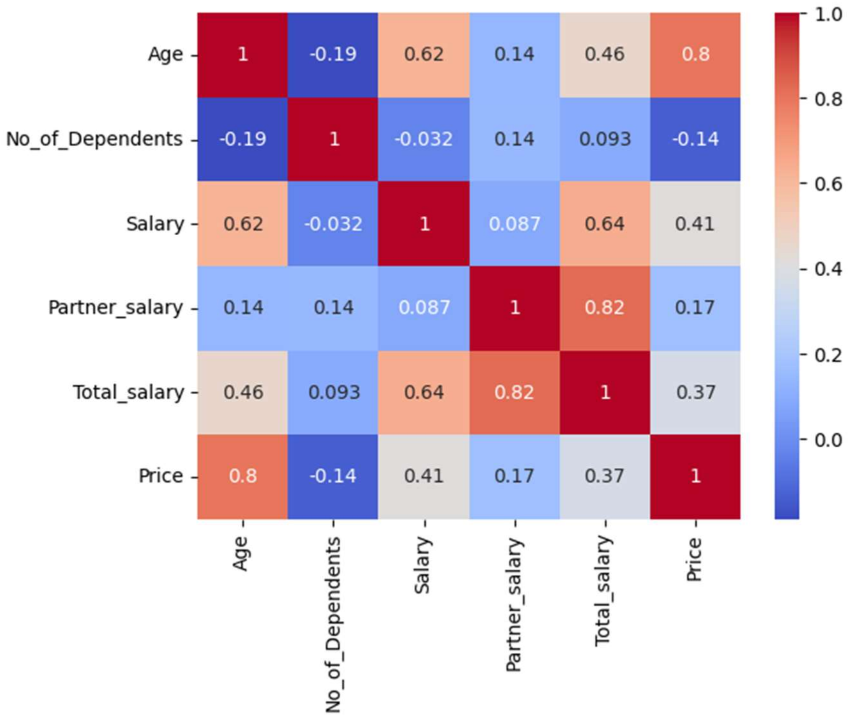
Bivariate Analysis

1. Between Numerical Values

- Exploring the relationship between all numerical variables



- Above pairplot shows relationship between all numerical values.



- Above Heatmap shows correlation between all numerical values

Positive Correlations

- **Car Price & Buyer Age:** There's a strong positive correlation (0.81) between the price of cars and the age of buyers, indicating that older individuals tend to purchase more expensive vehicles.
- **Total Salary & Partner's Salary:** These two variables also show a positive correlation, meaning that as one increases, the other tends to rise as well.

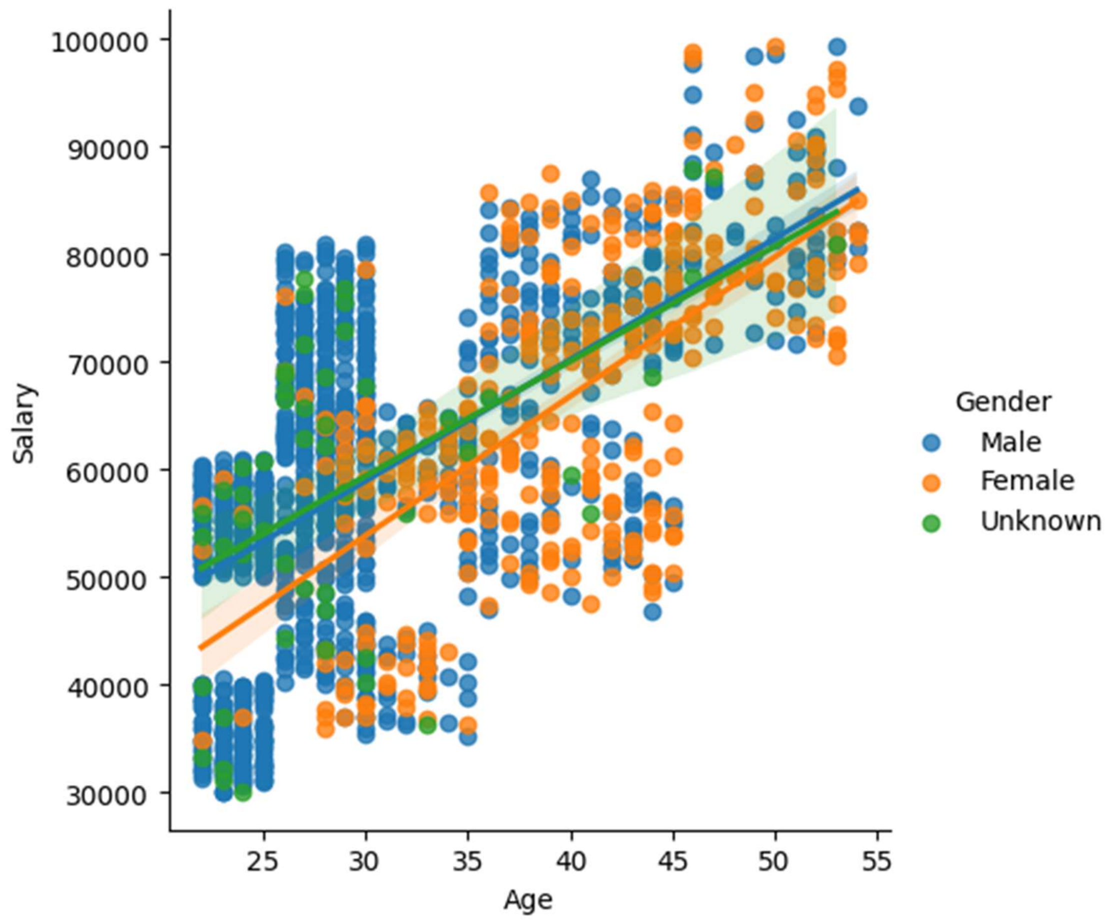
Negative Correlations

- **Number of Dependents vs Salary & Car Price:** A negative correlation exists between the number of dependents and both the buyer's salary and the price of cars. This suggests that as the number of dependents increases, total income and spending on cars tend to decrease.

2. Numerical and Categorical Variables

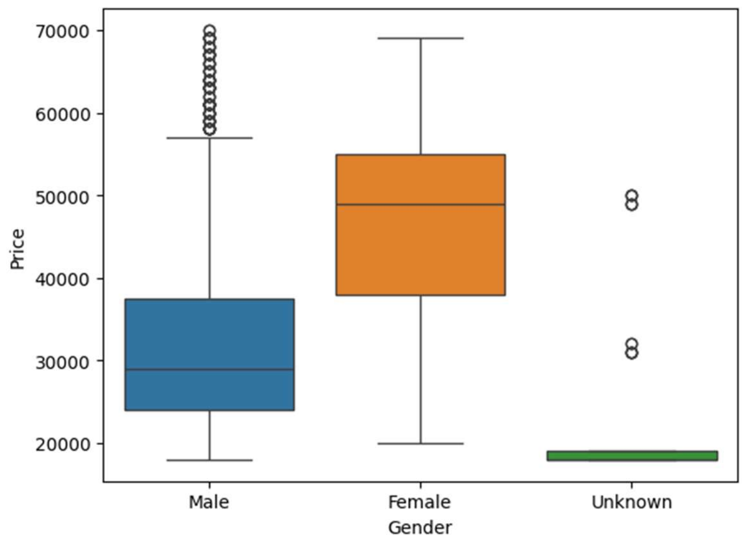
- Understanding the relationship between categorical and numerical variables

Age Vs Salary w.r.t Gender



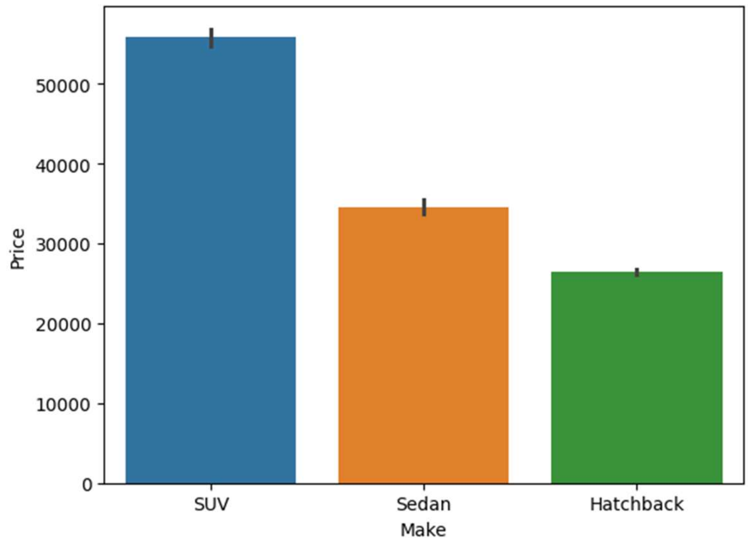
- Graph shows strong positive relation between age and salary, trend goes upward from left to right.
- Means when customer age is going higher their salary increases.

Gender Vs Price



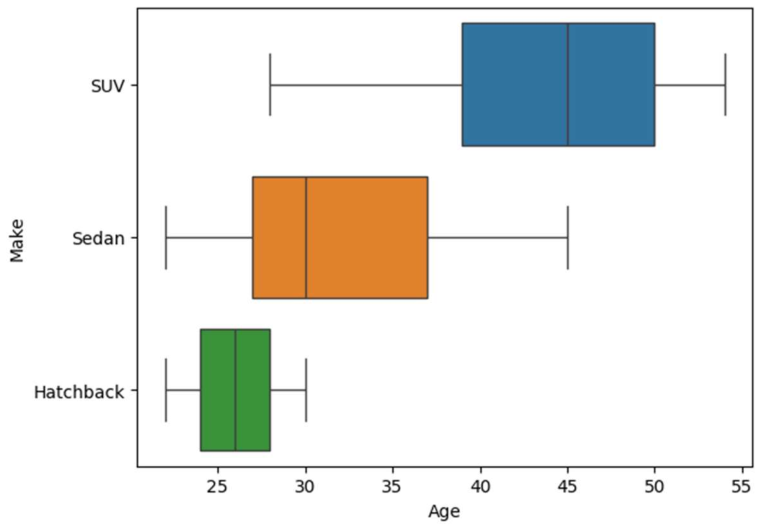
- **Female buyers** tend to purchase cars at **higher median prices** than male buyers.
- **Male buyers** show a wider spread in prices, with more **high-end outliers**.
- Buyers labelled as **'Unknown' gender** have the **lowest median price** and a very narrow price range.

Make Vs Price



- **SUVs** dominate in price, averaging just above $60,000, suggesting a premium market segment.
- **Sedans** sit in the middle range (~$40,000), likely appealing to mid-tier consumers.
- **Hatchbacks** are the most budget-friendly (~$30,000), ideal for cost-conscious buyers or urban commuters.

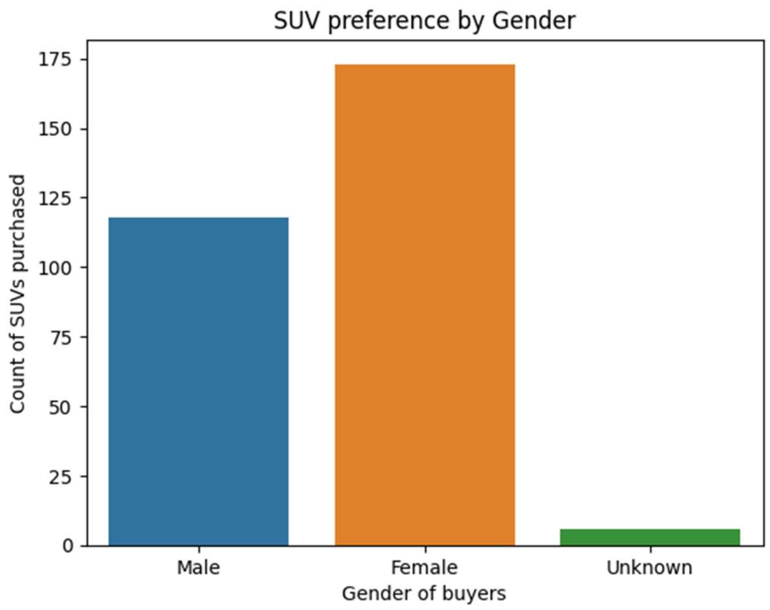
Age Vs Make



- SUV buyers tend to be older, with most around 45 years old.
- Sedan buyers are typically in their mid-30s.
- Hatchback buyers are the youngest group, mostly around 28 years old

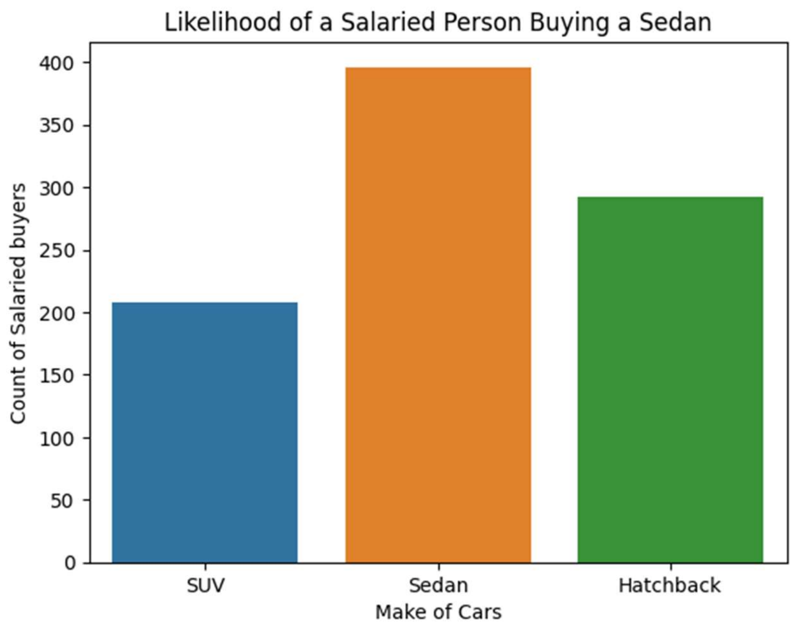
KEY QUESTIONS:

1. Do men tend to prefer SUVs more compared to women?



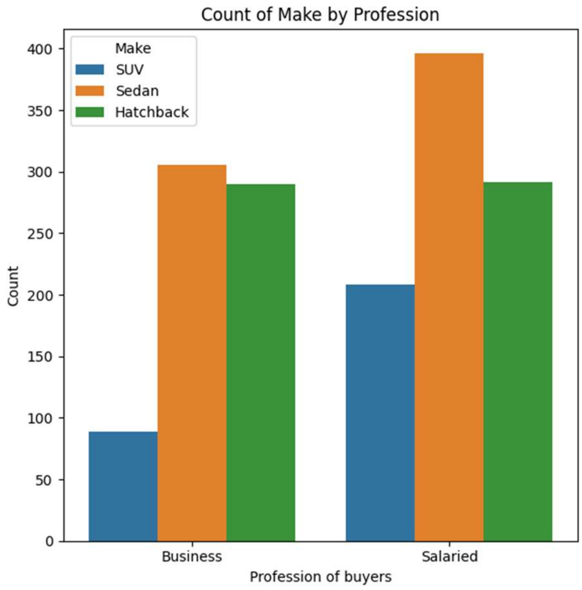
Women tend to show a **stronger** preference for purchasing **SUVs** compared to men, who are more inclined toward hatchbacks and sedans.

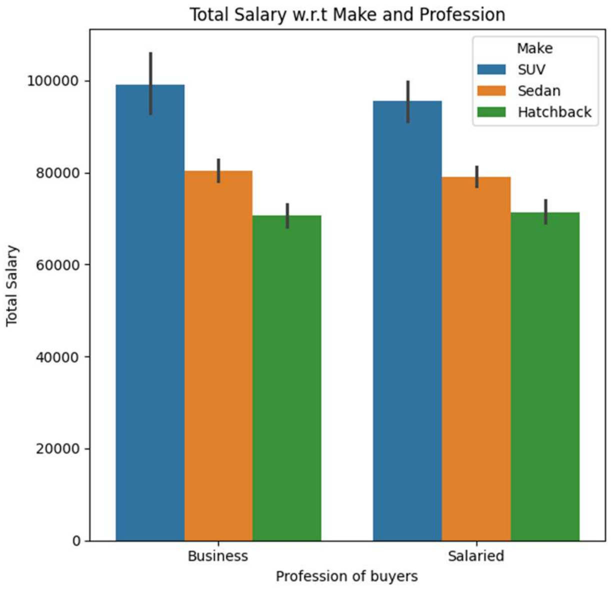
2. What is the likelihood of a salaried person buying a Sedan?



It can be concluded that salaried individuals are more likely to choose sedans over other car types.

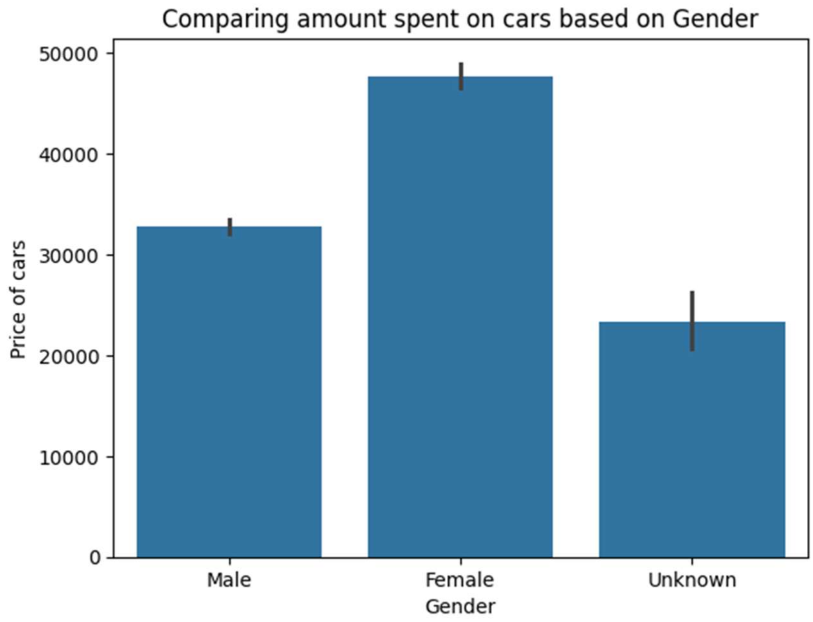
3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?





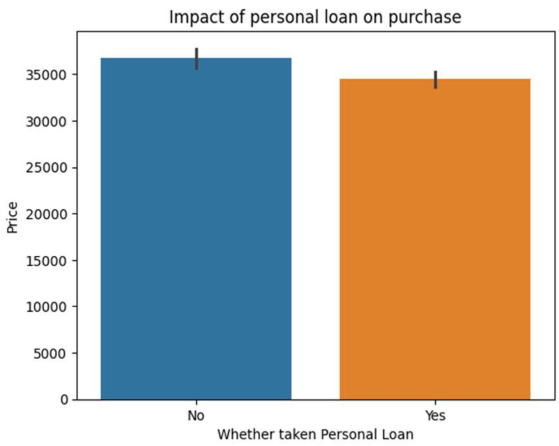
When it comes to car choices, salaried males seem to pick sedans over SUVs

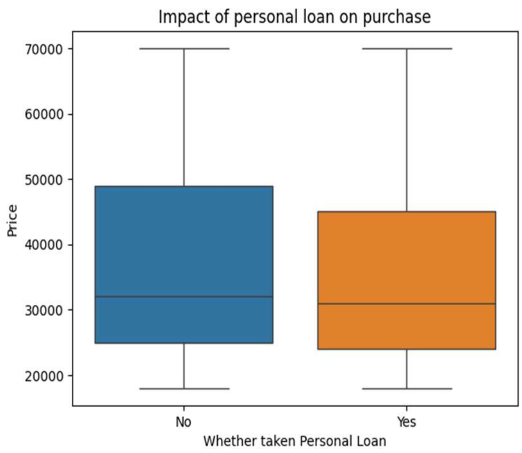
4. How does the amount spent on purchasing automobiles vary by gender?



- Overall, Females account for the highest total car expenditure, followed by Males, with the 'Unknown' category trailing far behind.

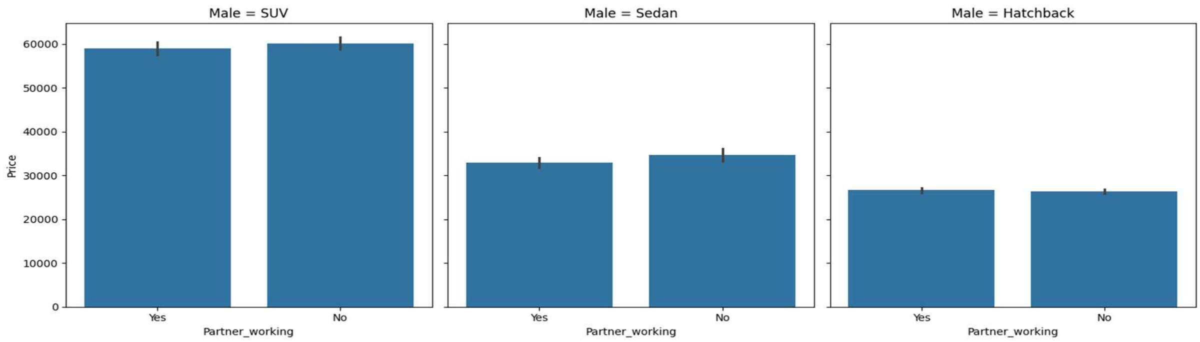
5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

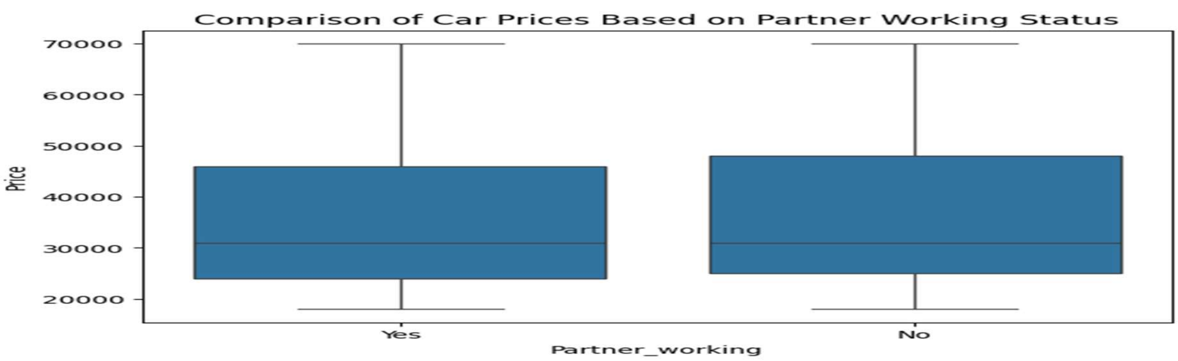




- **Average Price:** Non-loan buyers spent slightly more on average (over $37,000) than buyers with personal loans (under $37,000).
- **Typical Purchase:** The median price was nearly identical for both groups, showing that the middle-range car purchase isn't affected by the financing method.
- **Spending Habits:** Buyers with loans demonstrated more disciplined spending within a tighter price range. In contrast, those without loans had a much broader spectrum of spending.
- **Overall Takeaway:** Taking a personal loan seems to promote more conservative spending on a vehicle, rather than encouraging a more expensive purchase.

6. How does having a working partner influence the purchase of higher-priced cars?





- A non-working partner may lead to tighter household budgets, reducing the likelihood of buying higher-priced cars
- **Dual-income households** represent a prime target for **higher-end vehicles** like SUVs and premium Sedans.
- Marketing campaigns should emphasize **comfort, status, and long-term value** to appeal to these buyers.

Actionable Insights & Recommendations

Actionable Insights

1. **Ineffective "One-Size-Fits-All" Strategy**

Austo Motors' current marketing and sales approach is too generalized. It fails to resonate deeply because the customer base is not monolithic but composed of distinct segments—families, professionals, and young buyers—each driven by different needs and aspirations.

2. **Regional Demographics Drive Product Success**

There is a clear correlation between local demographics and sales performance. Dealerships in suburban, family-oriented areas see high demand for SUVs, while those in urban centers with younger populations sell more hatchbacks. This indicates that inventory and marketing require a localized approach.

3. **SUVs: The Engine of Growth Fueled by Financing**

The SUV segment is the primary revenue driver, particularly popular among family-focused consumers. A crucial factor enabling these sales is the widespread use of personal loans, making accessible and attractive financing a critical component of the buyer's journey.

4. **Hatchbacks: A Market Governed by Practicality and Price**

For the hatchback segment, purchasing decisions are overwhelmingly practical. Buyers are highly sensitive to price, prioritizing fuel economy, low maintenance, and overall affordability over performance or premium features.

Business Recommendations

1. **Adopt a Segment-Centric Go-to-Market Strategy**

Transform the sales and marketing functions to align with customer profiles.

- **Targeted Messaging:** Develop distinct campaigns for each vehicle type. Market SUVs with themes of safety and family; Sedans with sophistication and technology; and Hatchbacks with fun and economy.
- **Empower Sales Teams:** Train staff to quickly identify customer segments and tailor their sales pitch and vehicle recommendations to match the buyer's specific motivations.

2. **Optimize Inventory and Supply Chain for Local Markets**

Shift from a uniform inventory strategy to a data-driven, localized model.

- **Suburban Dealerships:** Increase stock of SUVs and family-oriented vehicles.
- **Urban & Youth-Heavy Areas:** Prioritize stocking a higher volume of hatchbacks and entry-level models.

3. **Build a Strategic Financial Partnership Program**

Acknowledge the importance of financing and use it as a competitive advantage.

- **Collaborate with Lenders:** Form partnerships with banks to create exclusive, low-interest loan packages, especially for the popular SUV models.
- **Promote Financing Options:** Integrate these financial solutions directly into marketing efforts to make the purchase proposition more attainable and appealing.

4. **Steer Product Innovation with Market Intelligence**

Use these insights to guide future planning and development.

- **Expand the SUV Portfolio:** Capitalize on strong demand by developing a wider range of SUV models at different sizes and price points.
- **Maintain Hatchback Competitiveness:** Ensure the hatchback lineup continues to lead on value by focusing on excellent mileage, affordability, and low ownership costs.